Direct Current (DC) Subtransmission Backbone for AI Factories

From Utility AC to 800 Vdc GPU Rails: Architecture, Efficiency, and Power Quality

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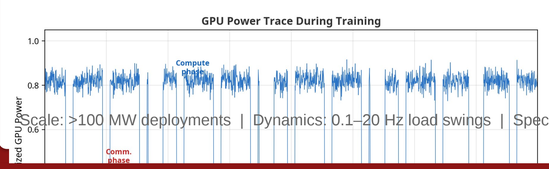
ABSTRACT & MOTIVATION

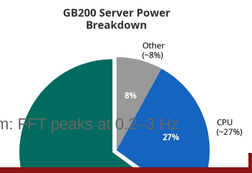
Why Revisit the Power Path Now?

AI-training loads are changing the problem from a purely steady-state efficiency problem into an architecture + power-dynamics problem. The GPU is the dominant power consumer (~65%), and synchronized GPU activity creates periodic power swings visible at the datacenter level.

Three Industry Scenarios:

1. Traditional AC: 5–6 conversion stages (Xfmr → UPS → PDU → PSU → VRM). Mature but inefficient.
2. AC-fed SST / 800 Vdc (NVIDIA et al.): Fewer stages, but AC still enters upstream — harmonics and PF issues persist.
3. Proposed MVDC hub: Single MV AC/DC front-end feeds MVDC backbone. Harmonics handled once. DC-native path for storage, solar, and rack buffering.





Scale: >100 MW deployments | Dynamics: 0.1–20 Hz load swings | Spectrum: FFT peaks at 0.2–3 Hz

PROPOSED ARCHITECTURE

Three Scenarios:

1. Traditional AC-centric

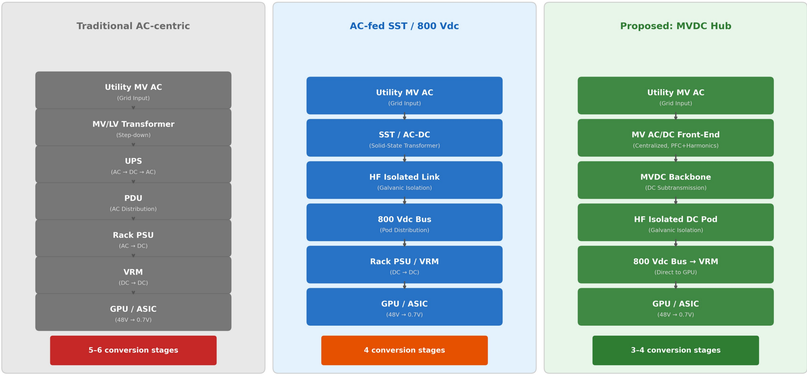
Utility MV AC → MV/LV Transformer → UPS (AC→DC→AC) → PDU → Rack PSU (AC→DC) → VRM (DC→DC) → GPU. Five to six conversion stages. Highest maturity but lowest efficiency due to repeated AC-DC-AC round-trips.

2. AC-fed SST / 800 Vdc Pod (NVIDIA et al.)

Utility MV AC → Solid-State Transformer (AC-DC) → HF Isolated Link → 800 Vdc Bus → Rack PSU/VRM → GPU. Four conversion stages. Reduces downstream complexity, but AC still enters at the pod front-end — each pod must independently handle harmonics and power factor correction.

3. Proposed: MVDC Hub + Isolated DC Pod

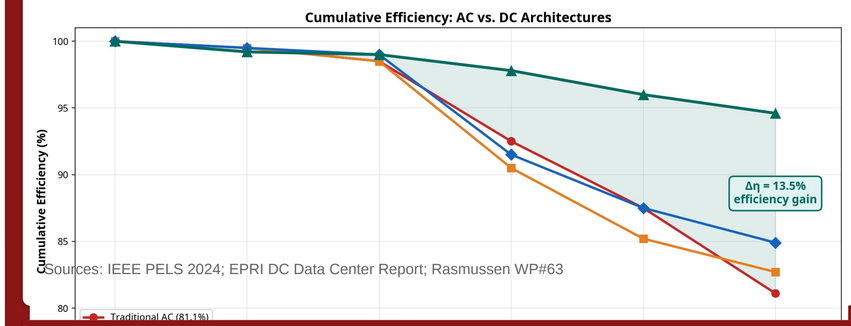
Utility MV AC → Centralized MV AC/DC Front-End (PFC + Harmonics) → MVDC Backbone (DC Subtransmission) → HF Isolated DC Pod → 800 Vdc Bus → VRM → GPU. Three to four conversion stages. Harmonics and PF are handled once at the centralized front-end. The DC backbone natively supports BESS, solar PV, and rack-level



BENEFIT 1: CUMULATIVE EFFICIENCY

The MVDC architecture eliminates redundant conversions, achieving 94.6% end-to-end efficiency vs. 81.1% for traditional AC — a 13.5 percentage-point gain.

For a 100 MW AI factory: ~13.5 MW saved ≈ $11.8M/year.



Sources: IEEE PELS 2024; EPRI DC Data Center Report; Rasmussen WP#63

BENEFIT 2: HARMONICS & POWER QUALITY

In AC architectures, every server PSU injects harmonics and requires power factor correction (PFC). With thousands of servers, harmonic management is distributed across 4–5 points in the power chain.

With MVDC, harmonic and PF handling become centralized once at the MV AC/DC front-end, instead of being repeatedly embedded in every downstream pod.

For utilities: AC-side power quality is managed only once at the grid interface — dramatically simplifying compliance with IEEE 519 harmonic limits and reducing filter equipment cost.

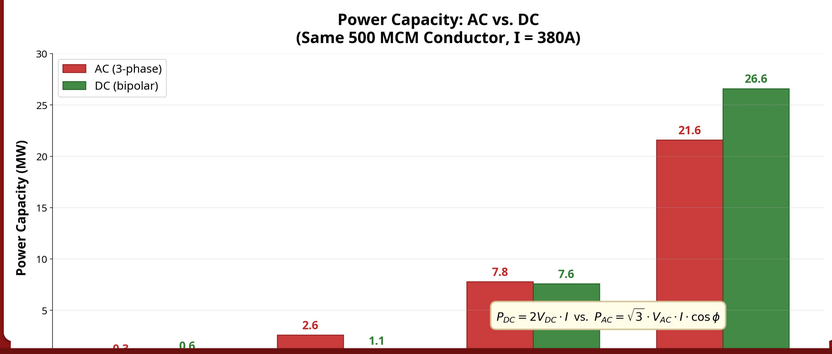
AC-centric: 4–5 distributed PQ management points
MVDC hub: 1 centralized PQ management point

BENEFIT 3: VOLTAGE MGMT & DC-NATIVE PATH

The MVDC backbone provides a cleaner DC-native path for rack buffering, battery storage (BESS), solar PV, and future modular AI power blocks — all connecting directly to the DC bus without AC-DC-AC re-conversion.

DC voltage regulation is inherently simpler: no reactive power, no frequency control, no phase balancing.

DC delivers ~28% more power per conductor at MV level.



ARCHITECTURE COMPARISON

The proposed path exchanges downstream simplicity for a tougher MVDC infrastructure problem.

Axis	Traditional AC	AC-fed SST 800 Vdc	MVDC Hub (Proposed)
Facility Interface	AC-native utility + facility protection	AC input still at pod / cluster front end	Centralized MV AC/DC feeding MVDC bus
Pod Complexity	Repeated downstream AC/DC + DC/DC stages	High: PFC + HF link + 800 Vdc output	Simpler: HF link only, AC/DC moves upstream
DC-Native Integration	Extra interfaces for storage / solar / DC	Better local DC, but AC still upstream	Best fit: BESS, solar, rack DC all native
Harmonic / PF Ownership	Distributed across AC facility + servers	Handled locally at each pod interface	Centralized at common MV front end
Maturity	Highest	Prototype to emerging products	Lowest today; protection is key hurdle
Main Bottleneck	Repeated conversions, copper, facility sprawl	SiC cost, HFT insulation, capacitor stress	MVDC protection, fault handling

The MVDC-hub proposal is strongest as a system-architecture story — not a claim that one converter solves the whole facility.

CONCLUSION & FUTURE WORK

Conclusion

The MVDC subtransmission backbone offers a compelling path for next-generation AI factories: fewer conversion stages, centralized power quality management, simpler voltage regulation, and native DC integration for storage and renewables.

Future Work:

- MVDC protection schemes, fault isolation, and safety standards development
- Techno-economic analysis at 100–500 MW scale with hardware-in-the-loop validation
- Optimal buffering placement (GPU, rack, pod, or MVDC hub) and BOM analysis

References

[1] IEA, "Data Centres and AI," 2024 [2] IEEE PELS, DC Distribution, 2024 [3] EPRI, DC Power Report, 2023
[4] NVIDIA, GB200 Power Architecture, 2024 [5] Rasmussen, APC WP#63 [6] Pratt et al., INTELEC 2007
[7] AlLee & Tschudi, IEEE Power & Energy, 2012